Maths 2

16 March 2026 20:24

- Check palindrome number
- Largest digit in a number
- Product of digits → HW
- Calculate a Prime Number
- Count number of divisors of a number
- Find GCD → HCF
- Find LCM using GCD

↳ $a \times b = GCD(a,b) \times LCM(a,b)$

Eg: 1 2 1 / 121

→ ←

↳ Reverse a number:

```
int n = ip();
int num = n;
int rev = 0;
```

→ reverse store (num)

```
while (num > 0) {
    int rem = num % 10;
    rev = rev × 10 + rem;
    num = num / 10;
}
```

reverse num

→ (num != 0)

1 2 1

3 → 32

3 × 10 + 2

10) 12 1 (12

120

1

```
// n → actual number
// rev → reversed number
if (rev == n)
    Sout("true");
else Sout("false");
```

Eg: n = 1 2 3 8 4 5 rem =

→ 3 2 1 8 6 6

0

```
int num = n;
int ans = 0;
while (num > 0) {
```

```
int ans = 0;
while (num > 0) {
  int rem = num % 10;
  if (rem > ans) {
      ans = rem;
  }
}
Sout (ans);
```

Q: 1 [2, 3, 4, 5 - - - - - - n]

2 → n-1

2, 3, 4 - - - - - , n-1 → true (prime)

→ prime →

5

5/2 → rem == 0 prime

true → false ✗

- 5/3 ×
- 5/4 ×

→ Prime

1) 2, 3, 4, 5 6

n=6

6/2 (rem == 0)

non-prime

6/3 ↳

→ int n = ip();

Boolean isPrime = true;

n = 5

i = 4 i = 5

```
Boolean isPrime = true;
for(int i=2; i<n; i++){
    if(n% i == 0) isPrime = false;
}
Sout(isPrime);
```

$i = 4$ $i = 5$

1) 2, 3, 4 (5

$2 \longrightarrow n-1$

$n/2$

$n/3$

$\sqrt{n}$

$n = 10^6$

$\sqrt{10^6}$

10^3

$n = 36$

$\frac{36}{2} = 18$

$1 \rightarrow 2 \times 18$

3×12

4×9

6×6

9×4

12×3

$6 \times 6 = 36$

$6^2 = 36 \rightarrow n$

$\sqrt{36} = 6$

$i <= \text{Sqrt}(n)$

$i <= \sqrt{n}$

$i \times i <= n$

36

1, 2, 3, 4, 5, 6, 7

$n = 36/2$

2, 3, 4, 5, 6, 7, 8 ---- 35

34 times

$4^2 = 16$

2×18

$n = 15$ int $m = \sqrt{15} = 3.$

$15/2 =$

$15/3 =$

$m = 3$

$i = 2$

$i = 3$

2, 3, 4, 5

3×5

5×3

6×6

3 5

5 3

for (i = 2 ; $i <= \sqrt{n}$; i++)

$i <= \sqrt{n}$

Sq. both Sides

$i^2 <= n$

$i \times i <= n$

Q. $n = 36$

→ 1, 2, 3, 4, 6, 9, 12, 18, 36

1, 2, 3, 4, 5 - - - - - - 36

36/1 ✓ Count = 1

36/2 Count = 2

Count = 9

```
int n = ip();
int ans = 0;
for (int i=1 ; i<=n ; i++){
    if (n%i ==0) ans++;
```

$\sqrt{n}$

1, 2, 3, - - - - - 36

36 / 1 = 36

```
if (n%i == 0)
}
sout(ans);
```

n = 15 → 1, 3, 5, 15

ans = 4

1, 2, 3,

1 × 15 → ans = 2

3 × 5 → ans = 4

36 = 36

1

i = 1 × 36

i = 2 × 18

i = 3 × 12 ans → 6

i = 4 × 9 ans → 7

i = 6 × 6 ans → 9

i = 9 × 4 →

ans = 4

$\sqrt{n}$

36

→ i = 2

i = 6

n/i

36/6 = 6

```
int n = i/p();
int ans=0;
for (int i=1; i*i<=n; i++) {
    if (n%i==0) {
        if (n/i==i) ans++;
        else ans+=2;
    }
sout(ans);
```